LEARNER GUIDE

Ten times bigger, ten times smaller

Place Value · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

When you multiply by 10, every digit moves **one column to the left**, so each is worth ten times as much. Multiplying by 100 moves the digits two columns, and by 1000 three columns. Dividing moves them to the **right**. The decimal point never moves. A 0 appears only to hold a column the digits have left empty: 36 × 10 = 360, because the 6 moved out of the ones column.

What you are learning to notice

- × 10, × 100, × 1000: the digits move 1, 2, 3 columns to the left.
- ÷ 10, ÷ 100, ÷ 1000: the digits move 1, 2, 3 columns to the right.
- The digits keep their order. Only their columns change.
- Zeros hold empty columns. They are not "added" on the end.
- To undo × 100, divide by 100: the digits move back.

Moving through it, one step at a time

Write the number in a place-value chart. Decide which way the digits move (left to multiply, right to divide) and how many columns (one for 10, two for 100, three for 1000). Move every digit that many columns, then fill any empty column up to the ones with a 0. Read the new number.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Multiplying by 10

What is 36 × 10, and where does the 0 come from?

36

Thousands	Hundreds	Tens	Ones
		3	6

36 × 10: every digit one column left

Thousands	Hundreds	Tens	Ones
	3	6	0

The highlighted 0 holds the ones column the 6 left empty.

1. 36 is	3 tens and 6 ones
2. × 10 moves each digit one column left.	3 hundreds and 6 tens
3. The ones column is now empty.	a 0 holds it
4. Read the new number.	360

36 × 10 = 360. The zero was not added: it holds the empty ones column.

Example 2 — Dividing by 100

What is 4,500 ÷ 100?

4,500

Thousands	Hundreds	Tens	Ones
4	5	0	0

4,500 ÷ 100: every digit two columns right

Thousands	Hundreds	Tens	Ones
		4	5

1. 4,500 is	4 thousands and 5 hundreds
2. ÷ 100 moves each digit two columns right.	4 tens and 5 ones
3. Check by undoing: × 100.	45 × 100 = 4,500

4,500 ÷ 100 = 45.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (what each column is worth (Lesson 1)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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